PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Effect of bortezomib on the treatment of multiple myeloma: A systematic review protocol
AUTHORS	Effect of bortezomib on the treatment of multiple myeloma: A systematic review protocol

VERSION 1 – REVIEW

REVIEWER	Nakamura, Kyohei QIMR Berghofer Medical Research Institute
REVIEW RETURNED	01-Mar-2022

GENERAL COMMENTS	This is a study protocol paper for a systematic review investigating effectiveness and safety of bortezomib in myeloma patients. I do not have specifically concens regarding study deisgn described in this masnucript.

REVIEWER	Derman, Benjamin University of Chicago Division of the Biological Sciences
REVIEW RETURNED	02-Mar-2022

GENERAL COMMENTS	- Under "Strengths and Limitations of this study" - should be 'data extraction' not excretion. - Page 5: ", with an estimated incidence in 185 countries" doesn't make sense. What is the incidence? - Page 5: 'Patients not eligible for transplantation receive 2-4 cycles'. This is not universally true. I would not be so dogmatic about myeloma treatment. You can say they are typically, or may be treated... - Page 6 " in the MM patients survivals, especially in developed countries" - should just be 'in survival' - Page 6: " thalidomide, bortezomib, and lenalidomide " - because thalidomide/lenalidomide are in same class I would put those together, or just say immunomodulatory imides (thalidoide/lenalidomide). - Page 6: "Additionally, in the last 8 years, several therapeutic options have been made available for elapsed patients, including carfilzomib, ixazomib, panobinostat, elotuzumab, pomalidomide, daratumumab, and selinexor." There is also elotuzumab and belantamab mafodotin. - page 7: "despite eligibility for ASCT" should be regardless of ASCT eligibility. - page 7: "The objective of this review is to evaluate the efficacy, effectiveness, and safety of bortezomib in the treatment of MM in patients over 18 years of age, eligible or not eligible for ASCT, and first-line or relapsed." try regardless of treatment setting and ASCT eligibility.

	- In regards to the technical aspects, it is not clear to me how uncontrolled trials will be used in this context? Will a network meta-analysis be required given the introduction of quadruplet therapies (ie. Rd vs VRd and VRd vs Dara-VRd)?

REVIEWER	Arcuri, Leonardo Javier Hospital Israelita Albert Einstein
REVIEW RETURNED	01-Apr-2022

GENERAL COMMENTS	Cantadori et al report the systematic review protocol for the effectiveness of bortezomib in the treatment of multiple myeloma. I have a few points: - The title seems wrong. Safety is a secondary outcome in the study. - The authors should choose one primary outcome for effectiveness. For multiple myeloma, I would choose progression-free survival - I'd like to understand the rationale of including observational studies because (1) there is more than one randomized trial available and (2) the potential bias is higher in observational studies. Please explain. - How can you see the sole effect of bortezomib in studies with different backbones? - Why will you use random effects model if you have not seen the heterogeneity? Will you use random effects model even with a low I-square? - The article is mainly written in the past tense. Please explain what has been already done (i.e., screening, full text selection, included studies)

VERSION 1 – AUTHOR RESPONSE

Reviewer 1
Thank you very much for reviewing our protocol.

Reviewer 2
Thank you very much for your comments and suggestions that absolutely improved the quality of our protocol. All your suggestions were included in the revised manuscript, as you can see below:

- Under "Strengths and Limitations of this study" - should be 'data extraction' not excretion.
Response: We apologize and corrected this mistake.

- Page 5: ", with an estimated incidence in 185 countries" doesn't make sense. What is the incidence?
Response: Again, we apologize and added the missing information. The right sentence should have been “According to the Global Cancer Observatory statistics, there were an estimated 160,000 cases of MM globally in 2018” as you now can see at page 5.
- Page 5: 'Patients not eligible for transplantation receive 2-4 cycles'. This is not universally true. I would not be so dogmatic about myeloma treatment. You can say they are typically, or may be treated...
Response: This is a great point in which we agree. Therefore, we made the changes as suggested.

- Page 6 " in the MM patients survivals, especially in developed countries" - should just be 'in survival'
Response: we apologize and corrected this mistake.

- Page 6: " thalidomide, bortezomib, and lenalidomide " - because thalidomide/lenalidomide are in same class I would put those together, or just say immunomodulatory imides (thalidoide/lenalidomide).
Response: Thank you for this addition. We made the changes as suggested.

- Page 6: "Additionally, in the last 8 years, several therapeutic options have been made available for elapsed patients, including carfilzomib, ixazomib, panobinostat, elotuzumab, pomalidomide, daratumumab, and selinexor." There is also elotuzumab and belantamab mafodotin.
Response: Thank you for this addition. We included the missing information.

- page 7: "despite eligibility for ASCT" should be regardless of ASCT eligibility.
Response: We apologize and corrected this mistake.

- page 7: "The objective of this review is to evaluate the efficacy, effectiveness, and safety of bortezomib in the treatment of MM in patients over 18 years of age, eligible or not eligible for ASCT, and first-line or relapsed." try regardless of treatment setting and ASCT eligibility.
Response: Thank you for this addition. We made the changes as suggested.

- In regards to the technical aspects, it is not clear to me how uncontrolled trialswill be used in this context? Will a network meta-analysis be required given the introduction of quadruplet therapies (ie. Rd vs VRd and VRd vs Dara-VRd)?
Response: We are doing this protocol in part on behalf of the Brazilian Health Ministry due to the recent incorporation of bortezomib into the public health system. Brazilian Health Ministry is interested in bortezomib efficacy, but also in its results from real world. Therefore, we were asked to include controlled observational studies, since the drug has been on the market for quite some time. It should be noted that observational studies will be evaluated separately from randomized clinical trials, and the GRADE system will be applied separately. In fact, the quality of the evidence generated will be different, and this will be highlighted.

Reviewer 3
Thank you very much for your comments and suggestions that absolutely improved the quality of our protocol. All your suggestions were included in the revised manuscript, as you can see below:

- The title seems wrong. Safety is a secondary outcome in the study.
Response: Thank you for the comment. Indeed, Safety is a secondary outcome in this study. Therefore, we changed the title as you suggested.

- The authors should choose one primary outcome for effectiveness. For multiple myeloma, I would choose progression-free survival
Response: Thank you for the comment. We agreed and proceeded as you suggested. Therefore, we chose progression-free survival as the sole primary outcome.

- I'd like to understand the rationale of including observational studies because (1) there is more than one randomized trial available and (2) the potential bias is higher in observational studies. Please explain.
Response: We are doing this protocol in part on behalf of the Brazilian Health Ministry due to the recent incorporation of bortezomib into the public health system. Brazilian Health Ministry is interested in bortezomib efficacy, but also in its results from real world. Therefore, we were asked to include controlled observational studies, since the drug has been on the market for quite some time. It should

be noted that observational studies will be evaluated separately from randomized clinical trials and the GRADE system will be applied separately. In fact, the quality of the evidence generated will be different, and this will be highlighted.

- How can you see the sole effect of bortezomib in studies with different backbones?
Response: Regarding the comparison between regimens with different backbones containing or not bortezomib and the comparison of bortezomib with other drugs: we make it more explicit on the protocol that the goal in this arm is to compare bortezomib with other drugs and regimens. Since myeloma treatment involves more than one antineoplastic agent concomitantly, we expect that this arm will have more complex comparisons. We won´t expect to be able to isolate the sole effect of bortezomib, but compare regimens which contain it with those which do not. Furthermore, this analysis will be performed separately from the arm of similar backbones with or without the addition of bortezomib.

- Why will you use random effects model if you have not seen the heterogeneity? Will you use random effects model even with a low I-square?

Response: We appreciate for the important question. According to the Cochrane handbook, the decision for fixed or random effects should be based on an expectation of whether the intervention effects are truly or unlikely identical among the studies plotted in the meta-analysis. The choice between a fixed-effect and a random-effects meta-analysis should never be made on the basis of the results of a statistical test for heterogeneity. Therefore, when studies are gathered from the published literature, studies that assess the impact of an intervention in different populations, the random-effects model is generally a more plausible match. Additionally, the random-effect methods and de fixed-effect method will give identical results when there is no heterogeneity among the studies

- The article is mainly written in the past tense. Please explain what has been already done (i.e., screening, full text selection, included studies)
we have done screening and we are now selecting the studies
Response: We have almost finished the selection process. Thus, in this revised version, we used simple past for search strategy, we have used present perfect for selection process and for other steps we have used the verbs in the future.

VERSION 2 – REVIEW

REVIEWER	Derman, Benjamin University of Chicago Division of the Biological Sciences
REVIEW RETURNED	06-May-2022

GENERAL COMMENTS	The authors have responded to my comments fully. No further suggestions. Good luck!

REVIEWER	Arcuri, Leonardo Javier Hospital Israelita Albert Einstein
REVIEW RETURNED	05-May-2022

GENERAL COMMENTS	The protocol carries a significant issue. The objective of the study is to evaluate the efficacy of bortezomib in the treatment of multiple myeloma, but their methodology hampers isolation the effect of bortezomib. Another issue is that the systematic review has already begun.

VERSION 2 – AUTHOR RESPONSE

Reviewer 3

The protocol carries a significant issue. The objective of the study is to evaluate the efficacy of bortezomib in the treatment of multiple myeloma, but their methodology hampers isolation the effect of bortezomib.

Response: In order to clarify our objectives, we have rewritten them, line 139 to 143: “This study aims to assess the effect of bortezomib on the OS, progression-free survival (PFS), overall response rate, time to next treatment, health-related quality of life, compliance, adverse events, and treatment-related death in patients with MM by comparing bortezomib treatment with the treatment without bortezomib in patients with the same background therapies, different background therapies, or other therapeutic agents.”. In order to clarify how our analyses will be performed, we have rewritten Intervention (I)/Comparator (C) section on line 162, as well as the measurement of treatment effect, line 246-251: “We will measure the effect of bortezomib in the treatment of MM in two analyses: (1) combining studies of bortezomib versus those without bortezomib in individuals with the same background therapy in each arm, and (2) combining studies of bortezomib versus those without bortezomib in individuals with different background therapies in each arm and studies of bortezomib versus those with other therapeutic agents”

Another issue is that the systematic review has already begun.

Response: Again, thank you very much for the important aspect that you have emphasized. Regarding protocol publication, we usually submit them to be published when they are in a stage before the extraction data process. Other scientific journals have also used this criterion, for example Systematic Reviews journal (“Protocol articles will only be considered for proposed or ongoing research that has not yet started the final data extraction stage of the review at the time of submission, and should provide a detailed account of the hypothesis, rationale and methodology of the study”). PROSPERO also accept registrations of protocol that are in the selection process(Submissions must be made before data extraction commences (from October 2019), https://www.crd.york.ac.uk/prospero/). In the BMJ open author guidelines, we have found the following statement: “We encourage the submission of protocol manuscripts at an early stage of the study. Protocols nearing completion of data collection will be treated on a case-by-case basis and the final decision on whether to consider a protocol for publication will rest with the Editor.” As we are in the selection process of our review, we truly believe that our protocol is in an early stage of the study. Additionally, we have added at the end of the protocolo the statement regarding study status: “Titles and abstracts have been screened by two independent reviewers for assessment against the inclusion criteria using the free web application Rayyan QCRI,” line 431

Again, thank you very much for your excellent contributions that have increased the methodological quality of our review.